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1. The cubic equation

$$x^3 - 5x^2 + px - q = 0$$

where p and q are positive real constants, has roots α , β and γ .

Given that

$$(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2 = 14$$

(a) show that $p = 6$

[3]

Given that

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 3$$

(b) determine the value of q

[3]

(c) Without solving the cubic equation, determine the value of $(\alpha + 1)(\beta + 1)(\gamma + 1)$

[4]

2. The quartic equation $x^4 + bx^3 + cx^2 + dx + 3 = 0$ has roots $\alpha, \beta, \gamma, \delta$. It is given that

$$\alpha + \beta + \gamma + \delta = -2, \quad (\alpha + 1)^2 + (\beta + 1)^2 + (\gamma + 1)^2 + (\delta + 1)^2 = 6, \quad \alpha^{-1} + \beta^{-1} + \gamma^{-1} + \delta^{-1} = 2$$

(a) Find the values of b, c and d [6]

(b) Given also that

$$(\alpha + 1)^3 + (\beta + 1)^3 + (\gamma + 1)^3 + (\delta + 1)^3 = 20$$

find the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$ [2]

- 3.** The equation $x^3 - 4x^2 - x + 6 = 0$ has roots α , β and γ .

Determine the value of

$$\frac{1}{\alpha + \beta} + \frac{1}{\beta + \gamma} + \frac{1}{\gamma + \alpha}$$

[4]

4. The quartic equation $3x^4 - x^3 - 10x^2 + 8x + 4 = 0$ has roots α , β , γ and δ .

Without solving the equation, find equations with integer coefficients whose roots are

(a) $\frac{1}{\alpha}$, $\frac{1}{\beta}$, $\frac{1}{\gamma}$ and $\frac{1}{\delta}$ [6]

(b) $\frac{2}{\alpha-1}$, $\frac{2}{\beta-1}$, $\frac{2}{\gamma-1}$ and $\frac{2}{\delta-1}$ [6]

5. The cubic equation $x^3 + x^2 - 2x - 1 = 0$ has roots α , β and γ .

(a) Find a cubic equation whose roots are $\frac{1}{\alpha+1}$, $\frac{1}{\beta+1}$ and $\frac{1}{\gamma+1}$ [3]

(b) Find the value of

$$\left(\frac{1}{\alpha+1}\right)^2 + \left(\frac{1}{\beta+1}\right)^2 + \left(\frac{1}{\gamma+1}\right)^2$$
 [2]

(c) Find the value of

$$\left(\frac{1}{\alpha+1}\right)^3 + \left(\frac{1}{\beta+1}\right)^3 + \left(\frac{1}{\gamma+1}\right)^3$$
 [2]

6. The equation $x^3 - 14x^2 + 56x + c = 0$, where c is a constant, has three roots in geometric progression.

(a) Determine the roots of the equation [6]

(b) Find c [1]

7. The quadratic equation

$$x^2 - kx + 1 = 0$$

where k is an integer, has roots α and β .

(a) Write down, in terms of k where appropriate, the value of $\alpha + \beta$ and the value of $\alpha\beta$ [2]

(b) Determine, in simplest form in terms of k , the value of

$$(\alpha^2 + \beta) + (\beta^2 + \alpha) \quad [3]$$

(c) Determine a quadratic equation which has roots

$$\alpha^2 + \beta \text{ and } \beta^2 + \alpha$$

giving your answer in the form $px^2 + qx + r = 0$, where p , q and r are integer values in terms of k [4]

8. The roots of the equation

$$x^3 + 5x^2 - 4x - 24 = 0$$

are α , β and γ .

Without solving the equation,

(a) write down the value of each of

$$\alpha + \beta + \gamma \quad \alpha\beta + \beta\gamma + \gamma\alpha \quad \alpha\beta\gamma \quad [1]$$

(b) Hence determine the value of

(i)

$$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} \quad [2]$$

(ii)

$$(\alpha + 2)(\beta + 2)(\gamma + 2) \quad [2]$$

(iii)

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 \quad [3]$$

9. The roots of the equation $3x^3 + px^2 - 12x - 8 = 0$ are α , β and γ .

(a) Given that $\alpha + \beta + \gamma = 2$, write down the value of p [1]

(b) Write down values for $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$ [1]

(c) Hence find the value of $\left(1 + \frac{1}{\alpha}\right) \left(1 + \frac{1}{\beta}\right) \left(1 + \frac{1}{\gamma}\right)$ [3]

10. The roots of the equation $x^3 + 2x^2 - 5x - 3 = 0$ are α , β and γ .

(a) Find $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$ [4]

(b) Find an equation with integer coefficients whose roots are α^2 , β^2 and γ^2 [4]

- 11.** The graph of $y = x^3 + 3x^2 + px - 20$, where p is a constant, crosses the x -axis at three points whose x -coordinates are equally spaced.

Find the roots of

$$x^3 + 3x^2 + px - 20 = 0$$

[6]

12. The cubic equation $x^3 - 3x + 1 = 0$ has roots α , β and γ .

(a) Find a cubic equation whose roots are $\frac{1}{\alpha-1}$, $\frac{1}{\beta-1}$, $\frac{1}{\gamma-1}$. [3]

(b) Hence find the value of $\frac{1}{(\alpha-1)^2} + \frac{1}{(\beta-1)^2} + \frac{1}{(\gamma-1)^2}$. [2]

(c) Find also the value of $\frac{1}{(\alpha-1)^5} + \frac{1}{(\beta-1)^5} + \frac{1}{(\gamma-1)^5}$. [3]

13. The quartic equation $x^4 - 2x^2 + 2x + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

(a) Find a quartic equation whose roots are $\alpha^2, \beta^2, \gamma^2, \delta^2$ and state the value of $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$. [5]

(b) Find the value of $\alpha^6 + \beta^6 + \gamma^6 + \delta^6$. [3]

(c) Find the value of $\alpha^8 + \beta^8 + \gamma^8 + \delta^8$. [2]

14. The equation $z^4 - 2z^3 - z^2 - 2z + 1 = 0$ has roots α, β, γ and δ .

(a) Show that a quartic equation whose roots are $\alpha + \frac{1}{\alpha}, \beta + \frac{1}{\beta}, \gamma + \frac{1}{\gamma}$ and $\delta + \frac{1}{\delta}$ is

$$w^4 - 4w^3 - 2w^2 + 12w + 9 = 0 \quad [3]$$

(b) Hence determine the exact roots of the equation $z^4 - 2z^3 - z^2 - 2z + 1 = 0$. [3]

15. The quartic equation $x^4 - 2x^3 + 3x^2 - 4x + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

(a) Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$. [2]

(b) Find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} + \frac{1}{\delta^2}$. [2]

(c) Find the value of

$$\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right)^2 + \left(\frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}\right)^2 + \left(\frac{1}{\gamma} + \frac{1}{\delta} + \frac{1}{\alpha}\right)^2 + \left(\frac{1}{\delta} + \frac{1}{\alpha} + \frac{1}{\beta}\right)^2$$
 [5]